EUNJEONG PARK

Ph.D student in Mechanical Engineering, The Pennsylvania State University
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RESEARCH INTERESTS

Additive Manufacturing (Metal 3D Printing), AI-driven Structural Optimization,
Large Language Models (LLM) for Scientific Discovery, Physics-Informed Machine Learning
Vision Transformer (ViT) for Microstructure Segmentation, Diffusion Models for Microstructure

EDUCATION

The Pennsylvania State University, United States *Aug.2025 - Present*

- Ph.D in Mechanical Engineering, GPA: 3.8/4.0
- Advisor: Prof. Amrita Basak
- University Graduate Fellowship

Korea Advanced Institute of Science and Technology, South Korea *Sep.2020 - Aug.2022*

- MS in Mechanical Engineering, GPA: 3.8/4.3
- Thesis: Machine Learning-based Elastic Modulus Prediction and Structural Shape Optimization
- Advisor: Prof. Seunghwa Ryu

Inha University, South Korea *Feb.2017 - Aug.2020*

- BS in Mechanical Engineering, GPA: 4.2/4.5
- Early Graduation, Dean's Scholarships and Hanjin Foundation Scholarship

WORK EXPERIENCE

Samsung Electronics *Aug.2022 - Jul.2025*

Semiconductor R&D Center - R&D Process Development Engineer (Dec.2024 - Jul.2025)

- Developed defect inspection process recipes for microscopy-based memory package products and led HBM4 equipment JEP ahead of mass production
- Conducted AFM JDP for 400mm next-gen advanced packaging and performed SAM simulations for void/delamination detection

Mechatronics Research - R&D Mechanical Engineer (Aug.2022 - Nov.2024)

- Conducted AFM & E-Beam JDP for metrology tool qualification and established inline 100% wafer inspection module
- Set up OCT-based full-wafer warpage measurement test bed; performed photoacoustic simulations for Cu void characterization in hybrid bonding structures

Inha University - 3D Printing & Laser Cutting Center *Mar.2020 - Jul.2020*

Student Assistant

- Assisted in the operation and maintenance of 3D printers and laser cutting machines
- Provided support to students and faculty on equipment usage and safety procedures

KIA Motors *Sep.2019 - Dec.2019*

Intern

- Contributed to an electric vehicle commercialization project, focusing on the development and implementation of a battery swapping system

PAPER

Multi-Label Phase Diagram Prediction in Complex Alloys via Physics-Informed Graph Attention Networks (*Under Review, May.2026*)

- **Eunjeong Park**, Amrita Basak

Free-form optimization of pattern shape for improving mechanical characteristics of a concentric tube (Materials & Design vol.230, *May.2023*)

- Hyunggwi Song†, **Eunjeong Park**†, Hong Jae Kim, Chung-Il Park, Taek-Soo Kim, Yoon Young Kim, Seunghwa Ryu

CONFERENCE

Machine Learning-based Pattern Shape Optimization to Improve the Mechanical Performance of Concentric Tube (KSME, *May.2022*)

- **Eunjeong Park**, Seunghwa Ryu

Transfer learning framework utilizing homogenization theory to predict the effective behavior of elasto-plastic composites (KSME, *May.2021*)

- Jiyoung Jeong, Yongtae Kim, **Eunjeong Park**, Seunghwa Ryu

HONORS AND AWARDS

University Graduate Fellowship (The Pennsylvania State University, *Aug.2025*)

Hanjin Foundation Scholarship (Hanjin Group, *Mar.2020*)

Super Challenge Hackerton, Best Award (Inha Univ. Start-up Center, *Feb.2020*)

University Simulation Competition, Honorable Mention (MSC software, *Feb.2019*)

3D Printing Creative Competition, Grand Prize (KSPSE, *Oct.2018*)

Comprehensive Design Competition, Grand Prize (Inha Univ. Innovation Center, *Sep.2018*)

LANGUAGES

Korean (Native) **English** (Working Proficiency)

ACADEMIC SKILLS

Programming Languages	Python, MATLAB
Simulation & Analysis	COMSOL Multiphysics, Ansys, ADAMS, SPPARKS, CALPHAD
CAD	Solid Edge, NX, AutoCAD
ML & AI	Machine Learning, Deep Learning, Gaussian Process, LLM, Vision Transformer (ViT), Diffusion Model
Fabrication & Optics	3D Printing (Metal/Polymer), Microscopy, AFM, e-Beam, WLI